

Sampling and Estimation

Complete detailed notes with formulas, examples, and solved explanations

Purpose of this PDF: to explain sampling, population vs sample, sampling techniques, resampling, sampling distribution, Central Limit Theorem, point estimation, interval estimation, confidence intervals, margin of error, and Z/t usage in a clear exam-ready format.

1. Inferential Statistics

Inferential statistics is used when we want to make conclusions about a large population using information from a smaller sample. In real life, studying the whole population is often costly, time-consuming, or impossible. So we select a representative sample, analyze it, and infer conclusions about the population.

Example: If a company has 1,00,000 customers, it may not survey every customer. Instead, it surveys 1,000 customers and uses the results to estimate the satisfaction level of all customers.

Descriptive Statistics	Inferential Statistics
Describes the data already available.	Uses sample data to make conclusions about a larger population.
Example: mean marks of 50 students.	Example: estimate average marks of all students using a sample.
Focus: summarize.	Focus: estimate, predict, test claims.

Main components of inferential statistics

- **Population:** the entire group we want to study.
- **Sample:** a smaller representative group selected from the population.
- **Estimation:** using sample statistics to estimate population parameters.
- **Confidence interval:** a range likely to contain the population parameter.
- **Hypothesis testing:** testing a claim about the population.
- **Regression analysis:** predicting relationships between variables.
- **ANOVA:** comparing means across multiple groups.

2. Population and Sample

Population

A population is the complete group of objects, people, observations, or events that we want to study.

Examples: all customers of a store, all workers in a factory, all students in a college, all patients in a hospital.

Sample

A sample is a smaller group selected from the population. The sample should represent the population properly so that our inference is reliable.

Population	Sample
Complete group of interest.	Subgroup selected from the population.
Population characteristic is called a parameter.	Sample characteristic is called a statistic.

Usually more accurate but expensive to study.	Cheaper and faster but has margin of error.
Example: all 10,000 students in a university.	Example: 500 students selected for survey.

Finite and Infinite Population

- **Finite population:** countable population. Example: number of workers in a factory.
- **Infinite population:** uncountable or practically impossible to count. Example: all stars in the Milky Way galaxy.

3. Parameter and Statistic

A parameter describes the whole population. A statistic describes a sample. In inferential statistics, we use sample statistics to estimate population parameters.

Concept	Symbol / Example	Meaning
Population mean	μ	Average of the whole population.
Population variance	σ^2	Variance of the whole population.
Population standard deviation	σ	Standard deviation of the whole population.
Sample mean	$\bar{x}$	Average of the sample.
Sample variance	s^2	Variance of the sample.
Sample standard deviation	s	Standard deviation of the sample.

Example: If we measure the length of 100 sparrows, the average length of those 100 sparrows is a statistic. We use it to estimate the average length of the entire sparrow population, which is a parameter.

4. Sampling

Sampling means selecting a smaller group from a population. We use sampling because studying the entire population may be difficult, costly, or impossible.

Main types of sampling

- Probability sampling and non-probability sampling.
- Univariate sampling and bivariate sampling.

5. Probability Sampling

Probability sampling is also called unbiased sampling. In this method, selection is random and each member of the population has a chance of being selected.

5.1 Simple Random Sampling

Every member of the population has an equal chance of selection.

Example: Selecting 100 employees randomly from 1,000 employees in a factory.

5.2 Systematic Sampling

Members are selected at regular intervals after choosing a starting point.

Example: Out of 1,000 employees, start from employee number 4 and select every 10th employee: 4, 14, 24, 34, and so on.

5.3 Stratified Sampling

The population is divided into subgroups called strata based on characteristics such as gender, age, income, or department. Then samples are selected from every stratum.

Example: From 700 males and 300 females, select 100 employees while maintaining the same balance. Select 70 males and 30 females.

5.4 Cluster Sampling

The population is divided into clusters. Some clusters are randomly selected, and all members from selected clusters are surveyed.

Example: A hospital groups patients by wards: Ward A, Ward B, Ward C. It randomly selects Ward A and Ward C and surveys all patients in those wards.

Stratified Sampling	Cluster Sampling
Population is divided into strata.	Population is divided into clusters.
Samples are taken from every stratum.	Entire selected clusters are surveyed.
Strata are usually different groups.	Clusters are usually mini-populations.
Example: select males and females proportionally.	Example: select full hospital wards.

Memory trick: Stratified = take some from all groups. Cluster = take all from selected groups.

6. Non-Probability Sampling

Non-probability sampling is also called biased sampling. Selection is non-random, so every member does not have an equal chance of being selected.

6.1 Convenience Sampling

Select people who are easiest to reach.

Example: asking your classmates or hostel friends to fill a survey.

6.2 Voluntary Response Sampling

People choose themselves to participate in the survey.

Example: an online poll where only interested users respond.

6.3 Judgement Sampling

The researcher or expert selects the sample based on judgment.

Example: judges selecting contestants in a reality show.

6.4 Snowball Sampling

First some people are selected. Then they help identify more people, and the sample grows like a snowball.

Example: while researching child labour, one child connects the researcher with others.

Probability Sampling	Non-Probability Sampling
Random selection.	Non-random selection.
Less biased.	More biased.
Every member has a chance.	Every member may not have a chance.

Better for generalization.

Useful when random sampling is difficult.

7. Univariate and Bivariate Sampling

Univariate Sampling

In univariate sampling, data is collected for only one variable.

Example: sample 100 people and record only their age.

It helps calculate mean, median, mode, variance, and standard deviation. It can be visualized using histogram, bar chart, pie chart, or box plot.

Bivariate Sampling

In bivariate sampling, data is collected for two related variables from each sample unit.

Example: sample 100 students and record study hours and exam score for each student.

Student	Study Hours	Exam Score
A	2	60
B	4	75
C	6	90

Bivariate data is used to study association, correlation, regression, and dependence between variables.

8. Resampling

Resampling means drawing samples again from an already available sample. It is useful for estimating uncertainty, standard error, and confidence intervals.

8.1 Bootstrap Resampling

Bootstrap resampling means taking random samples with replacement from the original sample. With replacement means the same value can appear more than once in the new sample.

Example: Original sample = 1, 4, 7, 2, 9, 3. A bootstrap sample may be 4, 4, 2, 9, 1, 1.

Bootstrap helps estimate standard error and confidence intervals for many statistics.

8.2 Jackknife Resampling

Jackknife resampling means repeatedly deleting observations from the original sample and recalculating the statistic.

Example: Original sample = 10, 20, 30, 40. Jackknife samples by deleting one observation are: 20, 30, 40; 10, 30, 40; 10, 20, 40; 10, 20, 30.

Bootstrap	Jackknife
Resample with replacement.	Delete observations and recompute.
Same observation can appear multiple times.	Each resample leaves out observation(s).
Useful for confidence intervals and standard error.	Useful for bias and variance estimation.

9. Sampling Distribution

A sampling distribution is the distribution of a sample statistic calculated from many samples drawn from the same population.

Example: Take many samples from a population. Calculate the sample mean for each sample. The distribution of all those sample means is called the sampling distribution of the sample mean.

Important characteristics

- **Mean:** the mean of the sampling distribution equals the population mean.
- **Standard error:** the standard deviation of the sampling distribution.
- **Shape:** for large sample sizes, the sampling distribution of sample mean becomes approximately normal.

$$\text{Standard Error for sample mean: } SE = \sigma / \sqrt{n}$$

Important: Larger sample size gives smaller standard error. This means estimates become more precise as sample size increases.

Example: Sampling Distribution of Sample Mean

Population: heights of all adults in a city. Population mean = 170 cm, population standard deviation = 10 cm. If many random samples of size 30 are taken, their sample means form a sampling distribution.

$$\text{Mean of sampling distribution} = \mu = 170$$

$$SE = \sigma / \sqrt{n} = 10 / \sqrt{30} = 1.83$$

Example: Sampling Distribution of Sample Proportion

Suppose 60% of voters support a candidate. Take random samples of 100 voters and calculate the supporter proportion in each sample.

$$\text{Mean of sample proportion distribution} = p = 0.6$$

$$SE = \sqrt{p(1-p)/n} = \sqrt{0.6(0.4)/100} = 0.049$$

10. Central Limit Theorem

Central Limit Theorem, or CLT, states that regardless of the shape of the original population, the sampling distribution of the sample mean approaches a normal distribution as sample size increases.

Common rule: sample size $n \geq 30$ is usually considered sufficient for CLT in many practical cases.

CLT is powerful because real-world data can be skewed, noisy, or non-normal. Even then, sample means tend to form a normal bell-shaped distribution when sample size is large.

Example: Daily sales of a coffee shop may be irregular. But if we repeatedly take samples and calculate their mean, the distribution of those means becomes approximately normal.

11. Statistical Inference

Statistical inference is the process of making conclusions about a population using sample information.

Example: Before crossing a road, you estimate the speed of an approaching car. Your estimate may not be exact, so you allow a margin of error. Similarly, statistical inference uses estimates with uncertainty.

Three main types

- Point estimation
- Interval estimation

- Hypothesis testing

12. Point Estimation

Point estimation uses sample data to calculate a single value that estimates an unknown population parameter.

Sample statistic	Estimates population parameter
Sample mean $\bar{x}$	Population mean μ
Sample proportion $\hat{p}$	Population proportion p
Sample standard deviation s	Population standard deviation σ

Sample mean: $\bar{x} = \sum(x) / n$

Sample proportion: $\hat{p} = x / n$

Example: Hospital Recovery Time

A hospital wants to estimate the average recovery time for all patients after surgery. It selects a sample of 50 patients and finds average recovery time = 8 days.

Point estimate: $\bar{x} = 8$ days. The hospital estimates that the average recovery time for all such patients is approximately 8 days.

Example: Point Estimate of Proportion

In a survey of 200 people, 80 said Yes, 100 said No, and 20 gave no response.

$$\hat{p}(\text{Yes}) = 80 / 200 = 0.4$$

$$\hat{p}(\text{No}) = 100 / 200 = 0.5$$

13. Interval Estimation

Interval estimation gives a range of values within which the population parameter is expected to lie. It is more informative than a point estimate because it includes uncertainty.

Example: Instead of saying average recovery time is exactly 8 days, we say it is between 7.45 and 8.55 days with 95% confidence.

Confidence Interval

A confidence interval is a range used to estimate a population parameter with a certain confidence level.

Confidence Level

The confidence level tells how confident we are that the interval contains the true population parameter. Common confidence levels are 90%, 95%, and 99%.

A 95% confidence level means that if the sampling process were repeated many times, about 95% of the calculated intervals would contain the true population parameter.

Margin of Error

Margin of error is the range above and below the sample statistic. It reflects uncertainty in the estimate.

$$\text{Estimate} \pm \text{Margin of Error}$$

Example: An exit poll says candidate X will get 40% votes with margin of error 5%. The interval is 35% to 45%.

14. Confidence Interval for Population Mean

Population mean estimate: $\mu = \bar{x} \pm \text{margin of error}$

If sigma is known: Margin of Error = $Z(\alpha/2) * \sigma / \sqrt{n}$

If sample standard deviation s is used: Margin of Error = $Z(\alpha/2) * s / \sqrt{n}$

Confidence Level	alpha	alpha/2	Z(alpha/2)
90%	0.10	0.05	1.645
95%	0.05	0.025	1.960
99%	0.01	0.005	2.576

Solved Example 1: Hospital Recovery Time

Given: sample size $n = 50$, sample mean $\bar{x} = 8$ days, sample standard deviation $s = 2$ days, confidence level = 95%, so $Z = 1.96$.

$$\text{Margin of Error} = 1.96 * 2 / \sqrt{50} = 0.55$$

$$\text{Confidence Interval} = 8 \pm 0.55 = (7.45, 8.55)$$

Interpretation: With 95% confidence, the true average recovery time for all patients lies between 7.45 and 8.55 days.

Solved Example 2: Boiling Temperature

A student records boiling temperature readings: 102.5, 101.7, 103.1, 100.9, 100.5, 102.2. Population standard deviation is known: $\sigma = 1.2$ C. Find 95% confidence interval.

$$n = 6, \sigma = 1.2, Z = 1.96$$

$$\bar{x} = (102.5 + 101.7 + 103.1 + 100.9 + 100.5 + 102.2) / 6 = 101.82$$

$$\text{Margin of Error} = 1.96 * 1.2 / \sqrt{6} = 0.96$$

$$\text{Confidence Interval} = 101.82 \pm 0.96 = (100.86, 102.78)$$

15. Confidence Interval for Population Proportion

Population proportion is estimated using sample proportion.

$$\hat{p} = x / n$$

$$\text{Confidence Interval for proportion: } p = \hat{p} \pm Z(\alpha/2) * \sqrt{\hat{p}(1-\hat{p})/n}$$

Solved Example: Yes Responses

A random sample of 400 individuals gives 100 Yes responses. Find the point estimate and 95% confidence interval.

$$\hat{p} = 100 / 400 = 0.25$$

$$Z = 1.96$$

$$\text{Margin of Error} = 1.96 * \sqrt{0.25(1-0.25)/400} = 0.0424$$

$$\text{Confidence Interval} = 0.25 \pm 0.0424 = (0.2076, 0.2924)$$

Interpretation: With 95% confidence, the true population proportion of Yes responses lies between 0.2076 and 0.2924, or 20.76% to 29.24%.

16. When to Use Z and When to Use t

Situation	Use
Large sample size, usually $n \geq 30$	Use Z-distribution
Population standard deviation σ is known	Use Z-distribution
Small sample size $n < 30$, population normal, σ unknown	Use t-distribution
Using t-distribution	Degrees of freedom = $n - 1$

Important exam rule: If $n < 30$, population is normal, and σ is unknown, use t-distribution with $n - 1$ degrees of freedom.

17. Complete Revision Table

Concept	Meaning
Inferential statistics	Uses sample to infer population.
Population	Entire group.
Sample	Smaller selected group.
Parameter	Population characteristic.
Statistic	Sample characteristic.
Probability sampling	Random and less biased.
Non-probability sampling	Non-random and more biased.
Simple random sampling	Every member has equal chance.
Systematic sampling	Select every k th member.
Stratified sampling	Sample from every subgroup.
Cluster sampling	Select entire clusters.
Bootstrap	Resample with replacement.
Jackknife	Delete observations and resample.
Sampling distribution	Distribution of sample statistics.
Standard error	SD of sampling distribution.
CLT	Sample means become normal for large n .
Point estimation	Single value estimate.
Interval estimation	Range estimate.
Confidence interval	Likely range for population parameter.
Margin of error	Uncertainty around estimate.
Z-distribution	Large sample or σ known.
t-distribution	Small sample and σ unknown.

18. Practice Questions

Use these questions to revise the chapter.

- Define population and sample with one example each.
- Differentiate between parameter and statistic.
- Explain simple random, systematic, stratified, and cluster sampling.
- Differentiate between stratified sampling and cluster sampling.
- Explain convenience, voluntary response, judgement, and snowball sampling.
- What is a sampling distribution?
- What is standard error? How does sample size affect it?
- Explain Central Limit Theorem with an example.
- Differentiate between point estimation and interval estimation.
- Calculate a 95% confidence interval when $\bar{x} = 50$, $s = 5$, $n = 100$.
- A sample of 500 people has 120 Yes responses. Find $\hat{p}$ and 95% confidence interval.
- When do we use t-distribution instead of Z-distribution?

19. Final Summary

Sampling and estimation help us understand a large population using a smaller sample. Sampling methods decide how the sample is selected. Probability sampling is more reliable because it is random, while non-probability sampling can be biased.

A parameter describes a population, and a statistic describes a sample. A sampling distribution shows how a statistic varies from sample to sample. Standard error measures the variation of a statistic across samples.

Central Limit Theorem explains why sample means become approximately normal when sample size is large. This allows us to build confidence intervals and perform inference even when the original population is not normal.

Point estimation gives one estimated value, while interval estimation gives a range. Confidence intervals use point estimate, margin of error, confidence level, and standard error to estimate population parameters.

This topic is important for surveys, polls, business analytics, hypothesis testing, machine learning evaluation, and real-world decision-making.